POLITECNICO
MILANO 1863

Clustering with covariates

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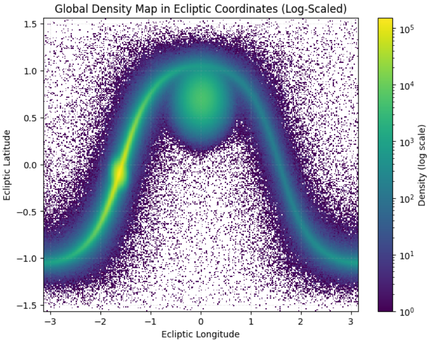
19/02/2026

Spatial Visualization:



Non-Uniform, clear curved pattern and high source concentration.

Laser Interferometer Space Antenna (LISA) Dataset:

- **Goal**: Identify and cluster double white dwarf (DWD) systems emitting gravitational waves.
- **Source**: $\approx$ **32M Observations** generated by the LISA mission.

Observed Variables:

- **Initial frequency (f_0)**
- **Frequency derivative ($\dot{f}$)**
- **Ecliptic latitude (b)**
- **Ecliptic longitude (l)**

Global Statistics

Statistic	f_0	$\dot{f}$	**b**	**l**
Mean	2.59×10^{-4}	6.19×10^{-17}	0.0065	-1.3682
Variance	5.85×10^{-8}	5.93×10^{-32}	0.1743	0.5450
Min	9.26×10^{-5}	7.97×10^{-23}	-1.5658	-3.1416
Max	3.15×10^{-2}	6.76×10^{-13}	1.5643	3.1416
Missing data	0			
Total observations	32,713,638			

Product Partition Model (PPM)

Data and notation:

- Observations: $\underline{y}_1, \ldots, \underline{y}_n$
- Cluster label: $c_i \in \{1, 2, \ldots\}$
- Partition: $\rho_n = \{S_1, \ldots, S_k\}$
- Prior over partition: $p(\rho_n)$
- Cluster size: $n_j = |S_j|$
- Cluster parameter: θ_j

1: Likelihood given the clustering

For each cluster S_j we assume:

$$y_i \mid c_i = j, \theta_j \overset{\text{ind}}{\sim} f(y_i \mid \theta_j), \qquad i \in S_j$$

2: The cluster parameters are a priori iid from the base measure P_0 :

$$\theta_j \overset{\text{iid}}{\sim} P_0.$$

3: Prior on the partition with a product structure:

$$p(\rho_n) \propto \prod_{j=1}^{k} c(S_j),$$

where $c(S_j)$ is the cohesion function of subset S_j.

Pitman–Yor Process

The Pitman–Yor (PY) process is a Bayesian nonparametric prior for random probability measures:

$$G \sim \mathrm{PY}(\alpha, d, G_0),$$

where:

- $\alpha > -d$ is the **strength** (or **concentration**) parameter,
- $0 \leq d < 1$ is the **discount** parameter,
- G_0 is the **base distribution** (prior mean of G).

Interpretation:

- α controls the overall probability of creating new clusters.
- d increases the preference for many small clusters and heavy–tailed cluster sizes.

Full conditional for cluster assignments

Allocated cluster:

$$P(c_i = j \mid \cdot) \propto (n_{-i,j} - d)\, \mathcal{N}(y_i \mid m_j, \sigma^2 + \tau_j^2)$$

where m_j and τ_j^2 are the posterior mean and variance of θ_j under a normal–normal update.

New cluster:

$$P(c_i = \text{new} \mid \cdot) \propto (\alpha + dk)\, \mathcal{N}(y_i \mid m_0, \sigma^2 + \tau_0^2)$$

PPMX: Product Partition Model with Covariates

Core Concept: The random partition ρ_n depends on **covariates** $x_{1:n}$, grouping observations with similar covariate profiles.

1. Partition Prior:

$$p(\rho_n \mid x_{1:n}) \propto \prod_{j=1}^{k} g(x_{S_j}) c(S_j)$$

2. Cluster Parameters:
Unique atoms drawn **i.i.d.**:

$$\theta_j \overset{\text{iid}}{\sim} G_0$$

3. Likelihood:
Conditionally **i.i.d.** given θ_j:

$$y_i \mid c_i = j, \theta_j \sim p(y_i \mid \theta_j)$$

The Full Joint Model:

$$P(y, \theta, \rho_n \mid x) \propto \underbrace{\prod_{j=1}^{k} \prod_{i \in S_j} p(y_i \mid \theta_j)}_{\text{Likelihood}} \times \underbrace{\prod_{j=1}^{k} p(\theta_j)}_{\text{Param Prior}} \times \underbrace{\prod_{j=1}^{k} g(x_{S_j}) c(S_j)}_{\text{PPMX Prior}}$$

The Similarity Function $g(x_{S_j})$

Marginal Likelihood Specification: We specify $g(\cdot)$ as the marginal likelihood of an auxiliary conjugate model:

$$g(x_{S_j}) = \int \left[\prod_{i \in S_j} q(x_i \mid \xi_j) \right] q(\xi_j) \, d\xi_j$$

Typical choices for $q(\cdot)$

- **Continuous**: Normal-Inverse-Gamma.
- **Categorical**: Dirichlet-Categorical.
- **Ordinal**: Ordinal Probit.
- **Count Data**: Poisson-Gamma.

Full Conditional for Updating c_i

Full Conditional Distribution:

$$P(c_i = j|\text{rest}) \propto P(c_i = j|c_{-i})p(y_i|y_{S_j \setminus \{i\}})\frac{g(x_{S_j \cup \{i\}})}{g(x_{S_j})}$$

Three Key Components

1. **Prior on the Partition** $(P(c_i = j|c_{-i}))$: Based on PPM / Pitman-Yor.
2. **Predictive Density** $(p(y_i|y_{S_j \setminus \{i\}}))$: Predictive density of y_i under cluster j.
3. **Covariate Similarity Adjustment** $\left(\frac{g(x_{S_j \cup \{i\}})}{g(x_{S_j})}\right)$: Adjustment based on covariate similarity.

Methodology I: Conjugacy & Marginalization

The implementation avoids the high-dimensional sampling of cluster parameters $\theta_j = (\mu_j, \Sigma_j)$ by utilizing **Conjugacy**.

- **Base Measure**: We use the Normal-Inverse-Wishart (NIW) as G_0.
- **Integration**: We analytically integrate out θ_j:

$$p(y_i \mid y_{S_j}, G_0) = \int f(y_i \mid \theta_j) p(\theta_j \mid y_{S_j}) d\theta_j$$

The "Collapsed" Advantage

By "collapsing" the sampler, the Markov Chain moves only through the space of **partitions** ρ_n, rather than parameters. This results in:

- Faster convergence (better mixing).
- Reduced memory overhead for large datasets like LISA.

Methodology II: The Collapsed Gibbs Step

The sequential Bayesian update of the NIW hyperparameters for each cluster S_j:

Posterior Hyperparameters:

- **Strength**: $\kappa_n = \kappa_0 + n_j$ and **Degrees of Freedom**: $\nu_n = \nu_0 + n_j$.
- **Location**: $m_n = \frac{\kappa_0 m_0 + n_j \bar{y}_j}{\kappa_n}$ (Shrinkage towards prior).
- **Scale**: $\Lambda_n = \Lambda_0 + S_j + \frac{\kappa_0 n_j}{\kappa_n}(\bar{y}_j - m_0)(\bar{y}_j - m_0)^T$.

Predictive Distribution

The result is a **Multivariate Student-**t distribution:

$$y_i \mid y_{S_j} \sim t_{\nu_n - D + 1}\left(m_n, \frac{\kappa_n + 1}{\kappa_n(\nu_n - D + 1)}\Lambda_n\right)$$

This is the exact likelihood used in the Gibbs transition probability.

Methodology III: Numerical Strategy

1. **Log-Space Arithmetic**: To avoid numerical underflow, we compute all weights in log-space.
2. **Mahalanobis Distance (D_M)**:

$$D_M^2 = (y_i - m_n)^T \Lambda_n^{-1} (y_i - m_n)$$

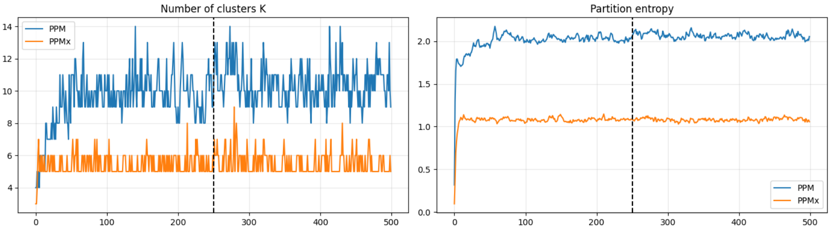
   This accounts for the **ellipsoidal shape** of clusters in the frequency-spatial domain.

PPMx Covariate Adjustment

$$\text{Prob} \propto \text{Prior} \times \text{Likelihood} \times \left[\frac{g(f_i, \dot{f}_i \cup x_{S_j})}{g(x_{S_j})} \right]^{\eta}$$

This ensures that sources with similar gravitational wave signatures $(f, \dot{f})$ are grouped together spatially.

Posterior Diagnostics: Number Of Clusters/Entropy



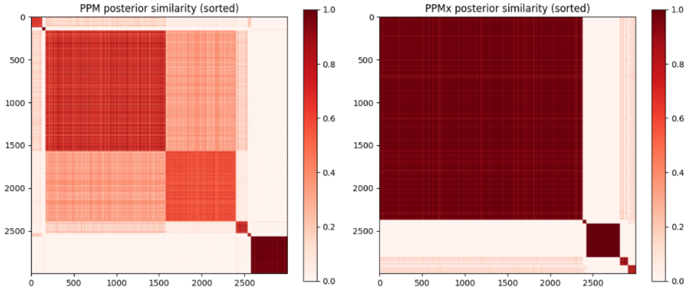
Number of Clusters K

	PPM	**PPMX**
Range	8–13	5–7
Variability	High	Low
Structure	Fragmented	Parsimonious

Partition Entropy

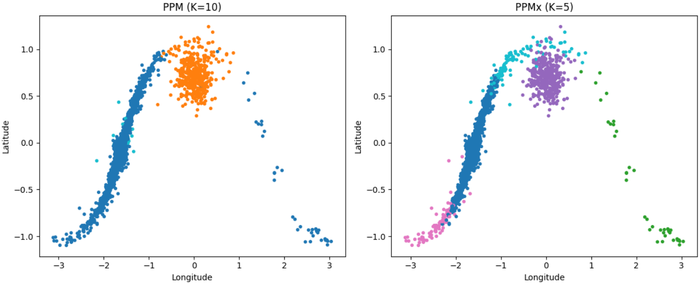
	PPM	**PPMX**
Entropy	~ 2.0	~ 1.1
Stability	Fluctuating	Stable
Structure	Fragmented	Coherent

Posterior Diagnostics: Posterior Similarity Matrix



	PPM	PPMX
Matrix structure	Diffuse / granular	Block-diagonal
Cluster boundaries	Weakly defined	Clearly separated
Co-clustering pattern	Intermediate values	Strong within clusters
Posterior uncertainty	Higher	Lower

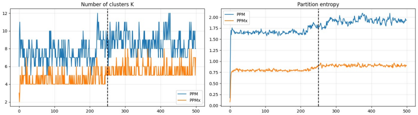
Posterior Diagnostics: Cluster Assignments



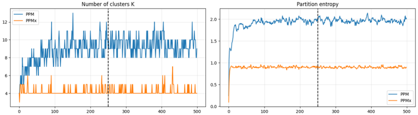
	PPM ($K = 10$)	**PPMX ($K = 5$)**
Clustering driver	Proximity	Proximity + covariates
Structure	Fragmented	More organized
Cluster boundaries	Irregular	Smoother / clearer
Posterior stability	Lower	Higher
Interpretability	Local density driven	Coherent groups

Prior Sensitivity: Variation of η

$\eta = 0.35$

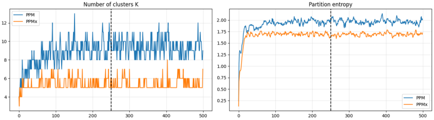


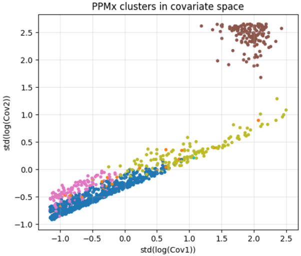
$\eta = 1$



$\eta = 0.35$	Weak covariate influence; higher variability in K and less smooth entropy trajectories.
$\eta = 1$	More concentrated posterior with smoother entropy and improved structural coherence.
Interpretation	Increasing η strengthens covariate contribution while preserving balance between response and covariate information.

Prior Sensitivity: Large $\eta = 1.5$

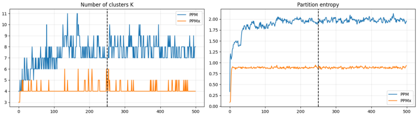




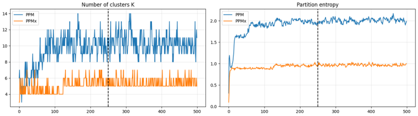
Entropy	Sharp increase and reduced stability, indicating higher posterior uncertainty.
Structure	Clusters align strongly with covariate density patterns.
Conclusion	η too large: clustering becomes overly conditioned by covariates and loses balance.

Prior Sensitivity: Variation of α

$\alpha = 0.5$

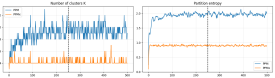


$\alpha = 2$

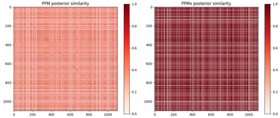


$\alpha = 0.5$	More parsimonious partition with fewer clusters and relatively stable dynamics.
$\alpha = 2$	Larger and more variable number of clusters, indicating stronger fragmentation.
Interpretation	Increasing α raises the tendency to create new clusters; very large values may over-fragment the true structure.

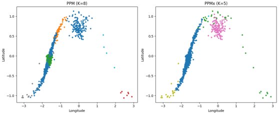
Prior Sensitivity: Selected Configuration $\alpha = 1$, $\eta = 1$



Entropy and K



Posterior similarity



Clustering result

- This configuration provides a balanced trade–off between structural clarity and model flexibility.
- Posterior similarity shows stronger agreement, with darker and more coherent blocks.
- Entropy remains low and stable, while the number of clusters avoids excessive fragmentation.

Conclusions

- **Bayesian Nonparametric Clustering with Covariates.** We applied a PPMX model to the LISA dataset to move beyond purely spatial clustering with different choices of hyperparameters. Compared to PPM, PPMX substantially improved the clustering results, producing more parsimonious and stable partitions, lower posterior uncertainty, and stronger physical coherence thanks to the inclusion of covariates. The Pitman–Yor prior captured heterogeneous cluster sizes and results remained robust.

- **Next Step: Clusters of Clusters.** The next goal is to build higher-level *meta-clusters* by grouping the inferred clusters. This second-stage analysis focuses on relationships between clusters rather than individual observations, aiming to reveal broader astrophysical structures and simplify global interpretation.

References

- **Barry, D. & Hartigan, J. A.** (1992)
  Product Partition Models for Change Point Problems.
  The Annals of Statistics, 20(1), 260–279.

- **Blanchard, P., Higham, D. J., & Higham, N. J.** (2021)
  Accurately Computing the Log-Sum-Exp and Softmax Functions.
  IMA Journal of Numerical Analysis, 41(4), 2311–2330.

- **Crowley, E. M.** (1997)
  Product Partition Models for Normal Means.
  Journal of the American Statistical Association, 92(437), 192–198.

- **De Blasi, P., Favaro, S., Lijoi, A., Mena, R. H., Prünster, I., & Ruggiero, M.** (2015)
  Are Gibbs-Type Priors the Most Natural Generalization of the Dirichlet Process?
  IEEE Transactions on Pattern Analysis and Machine Intelligence, 37(2), 212–224.

References

- **Greenwade, G. D.** (1993)
  The Comprehensive TeX Archive Network (CTAN).
  TUGBoat, 14(3), 342–351.

- **Mahalanobis, P. C.** (1936)
  On the Generalized Distance in Statistics.
  Proceedings of the National Institute of Sciences of India, 2, 49–55.

- **Müller, P., Quintana, F. A., & Rosner, G. L.** (2011)
  A Product Partition Model with Regression on Covariates.
  Journal of Computational and Graphical Statistics, 20(1), 260–278.

- **Murphy, K. P.** (2007)
  Conjugate Bayesian Analysis of the Gaussian Distribution.
  Technical Report, University of British Columbia.

- **Neal, R. M.** (2000)
  Markov Chain Sampling Methods for Dirichlet Process Mixture Models.
  Journal of Computational and Graphical Statistics, 9(2), 249–265.

- **Wade, S. & Ghahramani, Z.** (2018)
  Bayesian Cluster Analysis: Point Estimation and Credible Balls.
  Bayesian Analysis, 13(2), 559–626.

Thank you for your kind attention